Data Dictionary - Elec-Mart Retail and Ecommerce Performance Analytics

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1. Introduction

This project simulates a modern retail analytics platform for a mid-sized electronics company, designed to enable end-to-end decision-making across sales, marketing, inventory, and customer behavior.

A synthetic but realistic **3-Year dataset** was generated using Python (Faker, NumPy) and modeled using a modern ELT architecture (Snowflake + dbt). The platform supports scalable analytics use cases such as revenue optimization, customer segmentation, marketing attribution, and inventory planning. The dataset is composed of 14 interconnected tables stored in Parquet format, representing key business entities including customers, products, stores, promotions, campaigns, transactions, sales, inventory, and clickstream activity (Link to dataset).

Business logic and realistic constraints were embedded during the data generation process to mimic real operational scenarios. Detailed specifications of these rules are documented separately in the elecmart business-rules.md document.

This dataset enables end-to-end analysis across multiple domains, including sales performance, marketing effectiveness, online funnel analytics, inventory optimization, and customer behavior.

2. Data Lineage

This section outlines the end-to-end flow of data across the platform, from synthetic data generation to consumption in analytics and BI tools.

2.1. Data Flow Overview

This pipeline follows a modern ELT architecture, where raw data is loaded into the warehouse and transformed using dbt.

2.1.1. Stage Breakdown

1. Data Generation (Python)

- Synthetic data is generated using Python libraries such as Faker and NumPy
- Business rules and constraints are embedded to simulate realistic retail scenarios.
- Output is written as structured Parquet files.
- Future enhancement: implement SCD Type 2 for customer, product pricing, and store attributes to enable historical analysis.

2. Data Storage (S3 - Data Lake)

- Generated Parquet files are stored in Amazon S3
- Serves as the raw data lake layer

3. Bronze Layer (Snowflake)

- Raw data is ingested into Snowflake without transformation
- Tables mirror source structure (1:1 mapping from Parquet)
- Serves as the single source of truth for all downstream transformations

4. Silver Layer (dbt Transformations)

- Data is cleaned, standardized, and enriched using dbt models
- Key transformations include:
    - Handling missing and inconsistent values
    - Data type standardization
    - Removal of synthetic-only fields
- Ensures high-quality, analysis-ready datasets

5. Gold Layer (dbt Business Models)

- Data is transformed into business-ready models
- Star schema applied for performance and usability
- Includes:
  - Fact tables (sales, inventory, clickstream,transactions)
  - Denormalized dimension tables for BI consumption
- Data quality enforced through dbt tests and reconciliation checks

6. Consumption Layer (BI & Analytics)

- Gold layer tables are used in BI tools such as Tableau
- Supports dashboards and analysis across:
  - Sales performance
  - Customer behavior
  - Marketing effectiveness
  - Inventory optimization

3. Data Modeling Strategy

Relational modeling was used for the Bronze and Silver layers, and a star schema was applied in the Gold layer to support fast, accurate analysis and dashboarding.

3.1. Tables Overview - Bronze & Silver Layer

The table below highlights the high level description of all the tables in this project.

No.	Table Name	Table Description	Grain
1	(bronze/silver)_dim_date	Standard calendar dimension enabling time-based analysis across all datasets.	One row per calendar date

2	(bronze/silver)_dim_location	Centralized geographic reference supporting store, customer, and transaction analysis.	One row per unique location
3	(bronze/silver)_dim_store	Store master data including attributes such as type, size, location, and operational details.	One row per store
4	(bronze/silver)_dim_customer	Conformed customer dimension capturing demographics, acquisition, loyalty, and consent attributes.	One row per customer
5	(bronze/silver)_dim_brand	Reference table for product brands or manufacturers.	One row per brand
6	(bronze/silver)_dim_category	High-level product grouping for category-based analysis.	One row per category
7	(bronze/silver)_dim_subcategory	Product subcategories linked to parent categories.	One row per subcategory
8	(bronze/silver)_dim_product	Product master data including category hierarchy, brand, pricing, and warranty attributes.	One row per product (SKU level)
9	(bronze/silver)_dim_promotion	Reference table for promotions and discounts applied to transactions.	One row per promotion
10	(bronze/silver)_dim_campaign	Marketing campaign reference capturing channels, objectives, and timing.	One row per campaign
11	(bronze/silver)_fact_clickstream	Session-level data capturing user interactions such as views, add-to-cart, and purchases actions.	One row per session
12	(bronze/silver)_fact_sale	Line-level sales data capturing revenue, cost, and quantity for each product sold.	One row per transaction line item (product per

			transaction)
13	(bronze/silver)_fact_transaction	Transaction-level summary including totals, payment details, and transaction status.	One row per transaction
14	(bronze/silver)_fact_inventory	Snapshot of inventory levels across products and locations over time.	One row per product per location per month (snapshot grain)

3.1.1. Table Relationships

Table	**Relationship**
(bronze/silver)_dim_store → (bronze/silver)_dim_location	**Many-to-One via location_id**
(bronze/silver)_dim_customer → (bronze/silver)_dim_location	**Many-to-One via location_id**
(bronze/silver)_fact_transaction → (bronze/silver)_dim_customer	**Many-to-One via customer_id**
(bronze/silver)_fact_transaction → (bronze/silver)_dim_store	**Many-to-One via store_id**

(bronze/silver)_fact_transaction → (bronze/silver)_dim_promotion	**Many-to-One via promo_id**
(bronze/silver)_fact_clickstream → (bronze/silver)_fact_transaction	**One-to-Zero-or-One via session_id** ***(Optional relationship; applicable only for online transactions with valid session linkage)***
(bronze/silver)_fact_transaction → (bronze/silver)_dim_campaign	**Many-to-One via campaign_id**
(bronze/silver)_fact_sale → (bronze/silver)_fact_transaction	**Many-to-One via transaction_id**
(bronze/silver)_fact_sale → (bronze/silver)_dim_product	**Many-to-One via product_id**
(bronze/silver)_fact_inventory → (bronze/silver)_dim_store	**Many-to-One via store_id**
(bronze/silver)_fact_inventory → (bronze/silver)_dim_product	**Many-to-One via product_id**

(bronze/silver)_dim_product → (bronze/silver)_dim_category	**Many-to-One via category_id**
(bronze/silver)_dim_product → (bronze/silver)_dim_subcategory	**Many-to-One via subcategory_id**
(bronze/silver)_dim_product → (bronze/silver)_dim_brand	**Many-to-One via brand_id**
(bronze/silver)_dim_subcategory → (bronze/silver)_dim_category	**Many-to-One via category_id**
(bronze/silver)_dim_campaign → (bronze/silver)_dim_promotion	**Many-to-One via promo_id**
(bronze/silver)_fact_clickstream → (bronze/silver)_dim_customer	**Many-to-One via customer_id**
(bronze/silver)_fact_clickstream → (bronze/silver)_dim_campaign	**Many-to-One via campaign_id**
(bronze/silver)_dim_customer → (bronze/silver)_dim_date	**Many-to-One via signup_date_id**

(bronze/silver)_dim_store → (bronze/silver)_dim_date	**Many-to-One via opening_date_id**
(bronze/silver)_dim_campaign → (bronze/silver)_dim_date	**Many-to-One via campaign_start_date_id**
(bronze/silver)_dim_campaign → (bronze/silver)_dim_date	**Many-to-One via campaign_end_date_id**
(bronze/silver)_dim_promotion → (bronze/silver)_dim_date	**Many-to-One via promo_start_date_id**
(bronze/silver)_dim_promotion → (bronze/silver)_dim_date	**Many-to-One via promo_end_date_id**
All fact tables → (bronze/silver)_dim_date	**Many-to-One via their respective date_id fields i.e** **transaction_date_id, snapshot_month_id, session_start_date_id, session_date_id**

3.2. Table Overview - Gold Layer

The Gold layer is designed as the single source of truth for analytics and reporting. It applies a star schema to optimize query performance and usability in BI tools, enabling self-service analytics for business stakeholders.

No.	Table Name	Table Description	Grain
1	gold_fact_clickstream	Contains detailed user interaction data from the company's website, capturing events such as page views, session duration, and user behavior for digital analytics.	One row per session
2	gold_fact_sale	Contains line-level sales details for all transactions, including both completed and returned items, with product, quantity, cost, and pricing information. Additionally, this table was enriched with the allocated_line_discount and net_line_revenue columns.	One row per transaction line item
3	gold_fact_inventory	Provides aggregated inventory metrics by product, store, and month to support stock level monitoring and trend analysis.	One row per product, store, per month

4	gold_fact_transaction	Provides aggregated transactions details. Aggregates each sale by transaction id	One row per transaction
5	gold_dim_date	Provides a continuous calendar dimension to support time-based analysis, including attributes such as day, month, quarter, and year.	One row per calendar date
6	gold_dim_customer	Stores customer-level attributes used for segmentation and behavioral analysis, such as demographics, signup details, and customer persona.	One row per customer
7	gold_dim_store	Contains store-level details including store name, type, and associated location, supporting physical retail performance analysis.	One row per store
8	gold_dim_product	Holds product-level attributes such as product name, brand, category, and pricing information for sales and inventory analysis.	One row per product

9	gold_dim_campaign	Stores marketing campaign details such as campaign name, type, and duration for campaign performance analysis.	One row per campaign
10	gold_dim_promotion	Contains promotional and discount-related information applied to transactions, enabling analysis of promotion effectiveness.	One row per promotion

3.2.1. Table Relationships

Table	**Relationship**
gold_dim_store → gold_dim_date	**Many-to-One via opening_date_id**
gold_dim_campaign → gold_dim_date	**Many-to-One via campaign_start_date_id, campaign_end_date_id**
gold_dim_campaign → gold_dim_promotion	**Many-to-One via promo_id**

gold_dim_promotion → gold_dim_date	**Many-to-One via promo_start_date_id, promo_end_date_id**
gold_dim_customer → gold_dim_date	**Many-to-One via signup_date_id**
gold_fact_clickstream → gold_dim_date	**Many-to-One via session_start_date_id, session_end_date_id**
gold_fact_clickstream → gold_dim_campaign	**Many-to-One via campaign_id**
gold_fact_sale → gold_dim_date	**Many-to-One via transaction_date_id**
gold_fact_sale→ gold_dim_product	**Many-to-One via product_id**
gold_fact_inventory → gold_dim_date	**Many-to-One via snapshot_month_id**

gold_fact_inventory → gold_dim_product	**Many-to-One via product_id**
gold_fact_inventory → gold_dim_store	**Many-to-One via store_id**
gold_fact_transaction → gold_dim_customer	**Many-to-One via customer_id**
gold_fact_transaction → gold_dim_store	**Many-to-One via store_id**
gold_fact_transaction → gold_dim_promotion	**Many-to-One via promo_id**
gold_fact_transaction → gold_dim_campaign	**Many-to-One via campaign_id**
gold_fact_transaction → gold_dim_date	**Many-to-One via transaction_date_id**
gold_fact_sale → gold_dim_customer	**Many-to-One via customer_id**

gold_fact_sale → gold_dim_store	**Many-to-One via store_id**

3.3. Table Overview - Tableau Data Model

The Tableau semantic layer was intentionally designed to eliminate the need for data blending within Tableau. To achieve this, each Gold-layer fact table was configured as a separate Tableau data source, with only the relevant dimension tables joined to it. Each dashboard was then built using a single dedicated data source to ensure simplified modeling, improved query and filter performance, and consistent metric calculations. The data source utilization for dashboard creation is outlined below:

Gold Table/Data Source Name	**Dimension Connections**	**Used for**
gold_fact_inventory	• gold_dim_date via snapshot_month_id • gold_dim_product via product_id • gold_dim_store via store_id	Inventory Dashboard
gold_fact_transaction	• gold_dim_date via transaction_date_id • gold_dim_campaign via campaign_id • gold_dim_customer via customer_id • gold_dim_promotion via promo_id	Customer Dashboard
gold_fact_clickstream	• gold_dim_date via session_end_date_id • gold_dim_campaign via campaign_id • gold_dim_customer via customer_id	Clickstream Dashboard

gold_fact_sale	• gold_dim_date via transaction_date_id • gold_dim_product via product_id • gold_dim_store via store_id • gold_dim_campaign via campaign_id • gold_dim_promotion via promo_id	Executive Dashboard, Sales Dashboard, Campaigns Dashboard

4. List of Tables

4.1. DATES TABLE

Table Name: dim_date

Row Count: ~ 3,500 rows

Table Description: This is a date dimension table that provides a continuous calendar timeline for analytical queries.

Referenced by:

- dim_customer
- dim_store
- dim_campaign
- dim_promotion
- fact_clickstream
- fact_transaction
- fact_sale
- fact_inventory

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
date_id	Int (PK)	Surrogate key representing the calendar date in YYYYMMDD format	20250224	No	✅	✅	✅
date	Date	The full calendar date	2025-02-24	No	✅	✅	✅
year	Int	Calendar year	2025	No	✅	✅	✅
quarter	Int	Calendar quarter (1–4)	1	No	✅	✅	✅
month	Int	Month number (1–12)	2	No	✅	✅	✅
month_name	Varchar (50)	Full month name	February	No	✅	✅	✅

day	Int	Day of month (1–31)	24	No	✅	✅	✅
week_of_year	Int	Week number within the year	8	No	✅	✅	✅
day_of_week	Int	Day of week (0–6)	1	No	✅	✅	✅
day_name	Varchar(50)	Full weekday name	Monday	No	✅	✅	✅
is_weekend	Boolean	Indicates whether date falls on Saturday or Sunday	TRUE	No	✅	✅	✅
is_month_start	Boolean	Indicates if date is the first day of the month	TRUE	No	✅	✅	✅
is_month_end	Boolean	Indicates if date is the last day of the month	FALSE	No	✅	✅	✅

days_in_month	Int	Number of days in the month for the corresponding date	28	No	X	✅	✅
month_id	Int	The date value in ‘YYYYMM’ format	202508	No	X	X	✅
year_quarter	Varchar(50)	The year and quarter in format YYYY-Q	2025-Q3	No	X	X	✅

4.2. LOCATIONS TABLE

Table Name: dim_location

Row Count: ~ 25 rows

Table Description: Master reference for all geographic locations. Supports store, customer, and transaction-level analysis.

Referenced by:

- dim_customer
- dim_store

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Column Name	Data Type	Description	Example	Nullable	BL	SL	GL*
location_id	Int (PK)	Surrogate primary key uniquely identifying a geographic location at the country–state/province–city grain.	3	No	✅	✅	X
country	Varchar (50)	Standardized country name representing the top-level geographic entity for the location.	Canada	No	✅	✅	X
state_province	Varchar (50)	First-level administrative division of the country (e.g., state or province).	Ontario	No	✅	✅	X
city	Varchar (50)	City or municipality within the state or province.	London	No	✅	✅	X
location_type	Varchar (50)	This field was used for data generation and data enrichment	Urban	No	✅	✅	X

location_weight	Decimal(10, 2)	Relative weighting factor representing population density or commercial activity intensity for the location, primarily used for synthetic data generation and simulation.	0.25	No	✅	X	X
foot_traffic_min	Int	Lower bound of the expected foot traffic index range associated with the location, used to derive store-level foot traffic metrics.	80	No	✅	X	X
foot_traffic_max	Int	Upper bound of the expected foot traffic index range associated with the location, used to derive store-level foot traffic metrics.	99	No	✅	X	X

* Excluded from the gold layer because the gold layer adopts a star schema with all the location information populated in the dim_store and dim_customer tables.

4.3. CUSTOMERS TABLE

Table Name: dim_customer

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Row Count: 150,000 Customers

SCD Type: Type 0

Table Description: Conformed customer dimension including demographics, acquisition, loyalty, and marketing consent.

Referenced by:

- fact_clickstream
- fact_transaction

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
customer_id	Integer (PK)	System-generated unique identifier for each customer record	104552	No	✅	✅	✅
email_address	Varchar (50), Unique	Customer's email address, used as an alternate unique identifier	maria.thompson@xyz.com	No	✅	✅	✅
first_name	Varchar (50)	Customer's given (first) name	Maria	No	✅	✅	✅
last_name	Varchar (50)	Customer's family (last) name	Thompson	No	✅	✅	✅

gender	Varchar (50)	Self-reported gender of the customer	F	Yes	✅	✅	✅
customer_persona	Varchar (50)	Customer personas used to simulate purchasing behavior	Tech Enthusiast	No	✅	✅	✅
birth_date	Date	D.O.B of the customer	1991-01-26	No	✅	✅	X
birth_year	Int	Year of birth; customer must be at least 18 years old	1991	No	✅	✅	X
age	Int	The customer's age at the time of data generation	35	No	X	✅	X
age_group	Varchar (50)	The customer's age bucket	25 - 35	No	X	✅	✅
location_id	Int (FK references	Foreign key linking the customer to their primary location	3	No	✅	✅	X

	dim_location table)						
country	Varchar (50)	Denormalized into Gold layer for BI performance	Canada	No	X	X	✅
state_province	Varchar (50)	Denormalized into Gold layer for BI performance	Ontario	No	X	X	✅
city	Varchar (50)	Denormalized into Gold layer for BI performance	London	No	X	X	✅
location_type	Varchar (50)	Denormalized into Gold layer for BI performance	Urban	No	X	X	✅
signup_date	Date	Date the customer registered on the platform	2021-06-12	No	✅	✅	✅
signup_date_id	Int (FK references dim_dat	Foreign key linking the customer table to the date dimension table for analysis	20210612	No	✅	✅	✅

	e table)						
signup_channel	Varchar	Channel through which the customer was acquired (e.g., In-store, Web, Mobile App)	Online	Yes	✅	✅	✅
loyalty_status	Varchar	Current loyalty tier assigned to the customer	Gold	No	✅	✅	✅
estimated_annual_income	Decimal (10,2)	Estimated annual income of the customer	68,000	Yes	✅	X	X
income_bracket	Varchar	Grouping the customers by income to enable easier segmentation		Yes	X	✅	✅
email_opt_in	Boolean	Indicates whether the customer has consented to receive marketing emails	true	No	✅	✅	✅
sms_opt_in	Boolean	Indicates whether the customer has consented to	true	No	✅	✅	✅

		receive marketing SMS messages					

4.3.1. Data Quality Constraints

- signup_date <= CURRENT_DATE
- age >= 18

4.4. PRODUCTS TABLE

Table Name: dim_product
Row Count: 470 products

SCD Type: Type 0

Table Description: Master product reference, including category, subcategory, brand, pricing, and warranty. Supports product-level sales and margin analysis.

Referenced by:

- fact_sale

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
product_id	Int (PK)	Surrogate primary key uniquely	5501	No	✅	✅	✅

		identifying a product.					
product_name	Varchar (200)	Human-readable product name as listed by the retailer or distributor.	Samsung Galaxy S22	No	✅	✅	✅
category_id	Int (FK references dim_category table)	Foreign key referencing dim_category, representing the product's primary category.	3	No	✅	✅	X
subcategory_id	Int (FK references dim_subcategory table)	Foreign key referencing dim_subcategory, representing the product's subcategory.	15	No	✅	✅	X
brand_id	Int	Foreign key referencing dim_brand, representing the	15	No	✅	✅	X

	(FK references dim_brand table)	product manufacturer or brand.					
category_name	Varchar (50)	Denormalized into Gold layer for BI performance	Mobile	No	X	X	✅
subcategory_name	Varchar (50)	Denormalized into Gold layer for BI performance	Smartphones	No	X	X	✅
brand_name	Varchar (50)	Denormalized into Gold layer for BI performance	Samsung	No	X	X	✅
unit_cost	Decimal(10,2)	Unit acquisition cost incurred by the retailer for the product.	650.00	No	✅	✅	✅

unit_price	Decimal (10,2)	Standard unit selling price of the product to customers.	899.99	No	✅	✅	✅
warranty_years	Int	Length of manufacturer or retailer warranty in years. Null indicates no warranty coverage.	2	Yes	✅	✅	✅
product_segment	Varchar(50)	This field is used exclusively for synthetic data generation, feature engineering and categorisation	Flagship	No	✅	✅	✅

4.5. STORES TABLE

Table Name: dim_store
Row Count: 50 stores

SCD Type: Type 0

Table Description: Master reference for physical stores, including type, size, location, opening date, and foot traffic indicators.

Referenced by:

- fact_transaction

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
store_id	Int (PK)	Surrogate primary key uniquely identifying a store.	87	No	✅	✅	✅
store_name	Varchar, Unique	Human-readable store name as listed by the retailer or operator.	Elecmart Store #87	No	✅	✅	✅
location_id	Int (FK references dim_location table)	Foreign key referencing dim_location, identifying the geographic location of the store.	4	No	✅	✅	X

store_type	Varchar	Classification of store format or channel (e.g., Mall, Standalone, Outlet, Warehouse).	Mall	No	✅	✅	✅
store_size	Int	The physical floor area of the store measured in square meters.	500	No	✅	✅	✅
opening_date	Date	The date the store officially opened for business.	2014-09-15	No	✅	✅	✅
opening_date_id	Int (FK references dim_date table)	Foreign key linking the store table to the date dimension table for analysis	20140915	No	✅	✅	✅
foot_traffic_index	Int	Normalized indicator (0–100) representing expected customer	75	No	✅	✅	✅

		foot traffic for the store.					
location_type	Varchar (50)	Accepted values ('Rural', 'Urban', 'Suburban')	Urban	No	X	X	✅
country	Varchar (50)	Denormalized into Gold layer for BI performance	Canada	No	X	X	✅
state_province	Varchar (50)	Denormalized into Gold layer for BI performance	Ontario	No	X	X	✅
city	Varchar (50)	Denormalized into Gold layer for BI performance	London	No	X	X	✅

4.5.1. Data Quality Constraints

- opening_date < CURRENT_DATE - INTERVAL '2 Years'

4.6. CATEGORY TABLE

Table Name: dim_category

Row Count: ~ 10 Categories

SCD Type: Type 0

Table Description: High-level product categories for grouping products in analysis.

Referenced by:

- dim_product
- dim_subcategory

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
category_id	Int (PK)	Surrogate primary key uniquely identifying a product category.	1	No	✅	✅	X

category_name	Varchar (100)	Descriptive name of the product category.	Mobiles & Tablets	No	✅	✅	X

4.7. SUBCATEGORY TABLE

Table Name: dim_subcategory

Row Count: 28 subcategories

Table Description: Product subcategories linked to parent categories.

SCD Type: Type 0

Referenced by:

- dim_product

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
subcategory_id	Int (PK)	Surrogate primary key uniquely identifying a product subcategory.	10	No	✅	✅	X

subcategory_name	Varchar (100)	Descriptive name of the product subcategory.	Smartphones	No	✅	✅	X
category_id	Int (references dim_category table)	Foreign key referencing dim_category, associating the subcategory with its parent product category.	1	No	✅	✅	X

4.8. BRAND TABLE

Table Name: dim_brand

Row Count: ~ 50 brands

Table Description: Product brand or manufacturer master reference.

SCD Type: Type 0

Referenced by:

- dim_product

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
brand_id	Int (PK)	Surrogate primary key uniquely identifying a product brand.	16	No	✅	✅	X
brand_name	Varchar (100)	Descriptive name of the product brand.	Samsung	No	✅	✅	X

4.9. FACT TRANSACTION TABLE

Table Name: fact_transaction

Row Count:~ 900k Transactions (Over 3 Years)

Table Description: Transaction-level sales data.

Grain: one row per transaction.

Referenced by:

- fact_sale

Column	Data Type	Description	Example	Nullable	BL	SL	GL
transaction_id	Int (PK)	Surrogate primary key uniquely identifying a sales transaction.	10001	No	✅	✅	✅
transaction_timestamp	Timestamp	Timestamp at which the transaction was completed at checkout.	2024-05-12 14:32	No	✅	✅	✅
transaction_date_id	Int (FK references dim_date table)	Foreign key linking the transactions table to the date dimension table for analysis	20240512	No	✅	✅	✅
customer_id	Int (FK references dim_customer table)	Foreign key referencing dim_customer, identifying the customer who made the purchase. Null for anonymous	50214	Yes	✅	✅	✅

		purchases.					
store_id	Int (FK References dim_store table)	Foreign key referencing dim_store, identifying the physical or fulfillment store associated with the transaction.	87	No	✅	✅	✅
sales_channel	Varchar (30)	Channel through which the transaction was completed (e.g., In-store, Mobile, Web).	Mobile	No	✅	✅	✅
session_id	Int (FK references fact_clickstream table)	Foreign key referencing fact_clickstream, linking online transactions to the originating user session. Null for in-store purchases.	526	Yes	✅	✅	✅
promo_id	Int (FK references dim_pro	Foreign key referencing dim_promotion, identifying the	635	Yes	✅	✅	✅

	motion table)	promotion applied to the transaction. Null if no promotion was used.					
campaign_id	Int (FK references dim_campaign table)	Foreign key referencing dim_campaign, identifying the marketing campaign associated with the transaction. Null if no campaign is linked.	50	Yes	✅	✅	✅
transaction_subtotal	Decimal (10,2)	Total transaction value before discounts are applied.	900	No	✅	✅	✅
transaction_discount_applied	Decimal (10,2)	Total discount amount applied at the transaction (basket) level. Zero indicates no discount.	50	No	✅	✅	✅
transaction_total	Decimal (10,2)	Final transaction value after all discounts have been applied.	850	No	✅	✅	✅

transaction_cost	Decimal(10,2)	The total cost of all the products within this transaction	500	No	✅	✅	✅
items_count	Int	Total number of items purchased in the transaction.	3	No	✅	✅	✅
payment_type	Varchar (50)	Primary payment method used for the transaction (e.g., Credit, Debit, Wallet).	Credit	No	✅	✅	✅
transaction_status	Varchar (50)	Final status of the transaction (e.g., Completed, Returned).	Completed	No	✅	✅	✅

4.9.1. Data Quality Constraint

- transaction_total = transaction_subtotal - transaction_discount_applied
- transaction_timestamp < CURRENT_DATE
- transaction_total = (SELECT SUM(net_line_revenue) FROM gold_fact_sale WHERE gold_fact_sale.transaction_id = fact_transaction.transaction_id)
- transaction_subtotal = (SELECT SUM(line_total) FROM gold_fact_sale WHERE gold_fact_sale.transaction_id = fact_transaction.transaction_id)

- transaction_cost = (SELECT SUM(line_cost) FROM gold_fact_sale WHERE gold_fact_sale.transaction_id = fact_transaction.transaction_id)
- transaction_discount_applied = (SELECT SUM(allocated_line_discount) FROM gold_fact_sale WHERE gold_fact_sale.transaction_id = fact_transaction.transaction_id)

4.10. FACT SALE TABLE

Table Name: fact_sale
Row Count: ~1.8 million items (Over 3 Years)

Table Description: Item-level sales data.

Grain: one row per product line per transaction. Supports product-level revenue, discount allocation, and margin analysis.

Column Name	Type	Description	Example	Nullable	BL	SL	GL
sale_id	Int (PK)	Surrogate primary key uniquely identifying each sales line item.	2500	No	✅	✅	✅
transaction_id	Int (FK references the fact_tra	Foreign key linking the fact_sale table to the fact_transaction table	7821991	No	✅	✅	✅

	nsaction table)						
session_id	Int (FK references fact_clickstream table)	Foreign key linking the line item to the clickstream table for a completed transaction	693	Yes	✅	✅	✅
store_id	Int (FK references dim_store)	Foreign key linking the line item to the dim_store table	9	No	X	X	✅
customer_id	Int (FK references dim_customer)	Foreign key linking the line item to the dim_customer table	9637	Yes	X	X	✅
transaction_timestamp	timestamp	The timestamp the transaction was completed	2025-05-26 11:26:09	No	✅	✅	✅

transaction_date_id	Int (FK references dim_date table)	Foreign key linking the sale table to the date table	20250526	No	✅	✅	✅
product_id	Int (FK references dim_product table)	Foreign key identifying the product purchased.	5501	No	✅	✅	✅
quantity	Int	Number of units of the product purchased in the transaction. Must be a whole number ≥ 1.	2	No	✅	✅	✅
unit_cost	Decimal (10,2)	Unit acquisition cost of the product at the time of purchase.	100.00	No	✅	✅	✅
unit_price	Decimal (10,2)	Unit selling price of the product at the time of transaction, before	150.00	No	✅	✅	✅

		discounts.					
line_cost	Decimal (10,2)	Total cost for the line item (unit_cost × quantity).	200.00	No	✅	✅	✅
line_total	Decimal (10,2)	Gross line item value before discounts (unit_price × quantity).	300.00	No	✅	✅	✅
allocated_line_discount	Decimal (10,2)	The portion of the total transaction-level discount that is proportionally assigned to an individual line item (line_total / transaction_subtotal) * transaction_discount_applied	0.00	No	X	X	✅
net_line_revenue	Decimal (10,2)	The actual revenue attributed to a line item after deducting all applicable discounts.	0.00	No	X	X	✅

		(line_total - allocated_line_discount)					
transaction_status	Varchar (50)	Final status of the transaction (e.g., Completed, Returned).	Completed	No	X	X	✅

4.10.1. Data Quality Constraints

- transaction_timestamp < CURRENT_DATE
- quantity >= 1
- unit_cost < unit_price
- line_cost < line_total

4.11. INVENTORY TABLE

Table Name: fact_inventory
Row Count: ~846,000 (monthly records × 470 SKUs × 50 stores) - 3 Years

Table Description: Monthly inventory snapshots by store and product. Tracks starting stock, received stock, sold units, closing stock, backorders, and shrinkage.

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL

inventory_id	Int (PK)	Surrogate primary key uniquely identifying each inventory snapshot record.	889114	No	✅	✅	✅
store_id	Int (FK references dim_store table)	Foreign key referencing dim_store, identifying the store for this inventory snapshot.	87	No	✅	✅	✅
product_id	Int (FK references dim_product table)	Foreign key referencing dim_product, identifying the product for this inventory snapshot.	5501	No	✅	✅	✅
snapshot_month	Date	Month representing the inventory snapshot (first day of the month).	2024-03-01	No	✅	✅	✅

snapshot_month_id	Int (FK references dim_date table)	Foreign key linking the inventory table to the date dimension table	20240301	No	✅	✅	✅
starting_stock	Int	Quantity of product units available at the start of the snapshot month.	120	No	✅	✅	✅
received_stock	Int	Number of units received during the month.	50	No	✅	✅	✅
sold_units	Int	sold_units is derived from fact_sale by aggregating the total quantity sold per transaction per month	98	No	✅	✅	✅
closing_stock	Int	Quantity of product units remaining at the end of the snapshot month.	72	No	✅	✅	✅

backorder_flag	Boolean	Boolean indicating if any units were backordered during the month.	false	No	✅	✅	✅
shrinkage_loss	Int	Number of units lost due to shrinkage (theft, damage, etc.) during the month.	2	No	✅	✅	✅

4.12. MARKETING CAMPAIGNS TABLE

Table Name: dim_campaign
Row Count: ~ 120 rows (3 years)

SCD Type: Type 0

Table Description: Master data for marketing campaigns, including type, linked promotion, duration, and target segment.

Referenced by:

- fact_transaction

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL

campaign_id	Int (PK)	Surrogate primary key uniquely identifying the campaign.	301	No	✅	✅	✅
campaign_name	Varchar (200)	Descriptive name of the marketing campaign.	Holiday Email Blast	No	✅	✅	✅
campaign_channel	Varchar (50)	Type of campaign (e.g., Email, SMS, App Notification, Google Ads).	Email	No	✅	✅	✅
promo_id	Int(FK references dim_promotion table)	Foreign key referencing dim_promotion. Null indicates no promotion was linked.	531	Yes	✅	✅	✅
campaign_start_date	Date	Date the campaign began.	2023-12-01	No	✅	✅	✅

campaign_start_date_id	Int (FK references dim_date table)	Foreign key referencing dim_date table	20231201	No	✅	✅	✅
campaign_end_date	Date	Date the campaign ended.	2023-12-31	No	✅	✅	✅
campaign_end_date_id	Int (FK references dim_date table)	Foreign key referencing dim_date table	20231231	No	✅	✅	✅

4.12.1. Data Quality Constraints

- campaign_start_date <= campaign_end_date

4.13. FACT CLICKSTREAM TABLE

Table Name: fact_clickstream
Row Count: ~ 14 million rows (3 Years)

Table Description: Web session activity for anonymous and registered users. Grain: one row per session. Supports digital behavior and campaign attribution analysis.

Referenced by:

- fact_transaction

Column Name	Data Type	Description	Example	Nullable	BL	SL	GL
session_id	Int (PK)	Surrogate primary key uniquely identifying each web session.	112455998	No	✅	✅	✅
customer_id	Int (FK references dim_customer)	Foreign key referencing dim_customer. Null indicates anonymous or guest users.	104552	Yes	✅	✅	✅
session_start_time	Timestamp	Timestamp when the session began.	2024-01-05 10:22	No	✅	✅	✅
session_start_date_id	Int (FK references dim_date table)	Foreign key linking the clickstreams table to the date dimension table	20240105	No	✅	✅	✅

session_end_time	Timestamp	Timestamp when the session ended.	2024-01-05 10:56	No	✅	✅	✅
session_end_date_id	Int (FK references dim_date table)	Foreign key linking the clickstreams table to the date dimension table	20240105	No	✅	✅	✅
device_type	Varchar (50)	Type of device used for the session (e.g., Mobile, Tablet, Desktop).	Mobile	No	✅	✅	✅
number_of_pages_viewed	Int	Total number of pages the user visited during the session.	7	No	✅	✅	✅
product_page_visited_flag	Boolean	Boolean flag indicating whether the user visited any product page.	true	No	✅	✅	✅

added_to_cart_flag	Boolean	Boolean flag indicating whether the user added any product to their cart.	true	No	✅	✅	✅
purchased_flag	Boolean	Boolean flag indicating whether the session resulted in a completed purchase.	false	No	✅	✅	✅
traffic_source	Varchar (100)	Source of the session traffic (e.g., Organic, Paid, Email, Direct).	Paid	No	✅	✅	✅
linked_to_a_campaign_flag	Boolean	Boolean flag indicating whether the session is associated with a marketing campaign.	true	No	✅	✅	✅
campaign_id	Int (FK references dim_campaign table)	Foreign key referencing dim_campaign. Null indicates no campaign linkage.	1500	Yes	✅	✅	✅

aov_category	Varchar (50)	This field is used exclusively for synthetic data generation and feature engineering	Null	Yes	✅	X	X

4.13.1. Data Quality Constraints

- Session_end_time >= session_start_time

4.14. PROMOTIONS

Table Name: dim_promotion

Row Count: ~ 150 rows (3 years)

SCD Type: Type 0

Table Description: Master data for all promotions applied to campaigns and transactions. Includes discount type, value, duration, code, and description.

Referenced by:

- dim_campaign
- fact_transaction

Column Name	Type	Description	Example	Nullable	BL	SL	GL
promo_id	Int (PK)	Surrogate primary key uniquely identifying each promotion.	123	No	✅	✅	✅
promo_name	Varchar(100)	Human-readable name of the promotion.	Black Friday 20% Off	No	✅	✅	✅
promo_type	Varchar(50)	Classification of the promotion based on scale or impact (e.g., Major, Medium, Flash).	Major	No	✅	✅	✅
discount_type	Varchar(50)	Type of discount applied: percentage_discount or fixed_amount_discount.	percentage_discount	No	✅	✅	✅

discount_value	Decimal(10,2)	Value of the discount. Interpreted as a percentage or fixed amount depending on discount_type.	20.00 (%)	No	✅	✅	✅
promo_start_date	Timestamp	Date and time when the promotion becomes active.	2025-11-01 00:00:00	No	✅	✅	✅
promo_start_date_id	Int (FK references dim_date table)	Foreign key linking the promotions table to the date dimension table	20251101	No	✅	✅	✅
promo_end_date	Timestamp	Date and time when the promotion expires.	2025-11-15 23:59:59	No	✅	✅	✅
promo_end_date_id	Int (FK references dim_date table)	Foreign key linking the promotions table to the date dimension table	20251115	No	✅	✅	✅

promo_duration	Int	Duration of the promotion in days.	14	No	✅	X	X
promo_code	Varchar(100)	Promotion code used by customers at checkout.	SAVE20	No	✅	X	X
is_active	Boolean	Indicates if the promotion is active at the time of data generation. This is a point in time field and not a live indicator.	false	No	✅	✅	✅
promo_description	Varchar	Detailed explanation of the promotion.	20% off all items during Black Friday.	No	✅	X	X

4.14.1. Data Quality Constraints

- promo_start_date <= promo_end_date

5. Business Metrics Definition

5.1. Revenue & Profitability

Metric	Business Definition	SQL Logic	Grain	Notes
Gross Revenue	Total revenue after discounts before accounting for returns	`SUM(net_line_revenue)`	gold_fact_sale	Includes both completed and returned transactions
Net Revenue	Revenue from successfully completed transactions	`SUM(net_line_revenue) WHERE transaction_status = 'Completed'`	gold_fact_sale	Primary revenue KPI
COGS	Cost of goods sold for completed transactions	`SUM(line_cost) WHERE transaction_status = 'Completed'`	gold_fact_sale	Excludes returned transactions
Gross Profit	Profit after cost of goods sold	`Net Revenue - COGS`	gold_fact_sale	Core profitability metric
Gross Margin %	Profitability as a percentage of revenue	`Gross Profit / Net Revenue`	gold_fact_sale	Key margin indicator

5.2. Sales Performance

Metric	Business Definition	SQL Logic	Grain	Notes

Total Transactions	Number of completed purchase events	`COUNT(DISTINCT transaction_id) WHERE transaction_status = 'Completed'`	Gold_Fact_Sale	Excludes returned transactions
Returned Transactions	Number of returned purchase events	`COUNT(DISTINCT transaction_id) WHERE transaction_status = 'Returned'`	Gold_Fact_Sale	Excludes completed transactions
Total Orders	All transactions regardless of outcome	`COUNT(DISTINCT transaction_id)`	Gold_Fact_Sale	Includes returned transactions
Units Sold	Total quantity of items successfully sold	`SUM(quantity) WHERE transaction_status = 'Completed'`	Gold_Fact_Sale	Net of returns
Units Returned	Total quantity of returned items	`SUM(quantity) WHERE transaction_status = 'Returned'`	Gold_Fact_Sale	Helps track product issues
Average Order Value (AOV)	Average revenue per completed transaction	`Net Revenue / Total Transactions`	Gold_Fact_Sale	Based on completed orders only

5.3. Return Metrics

Metric	Business Definition	SQL Logic	Grain	Notes
Returned Revenue	Value of transactions that were returned	`SUM(net_line_revenue) WHERE`	Gold_Fact_Sale	Derived from sales table

		`transaction_status = 'Returned'`		
Return Rate %	Percentage of revenue that was returned	`Returned Revenue / Gross Revenue`	Gold_Fact_Sale	Indicates product or service issues
Return Transaction Rate %	Percentage of transactions that were returned	`Returned Transactions / Total Orders`	Gold_Fact_Sale	Operational quality metric

5.4. Inventory Metrics

Metric	Business Definition	SQL Logic	Grain	Notes
Inventory Turnover	Stock efficiency	SUM(sold_units) / AVG((starting_stock + closing_stock) / 2.0)	Monthly from Gold_Fact_Inventory	Performance metric
Total Shrinkage Loss	Value of inventory lost to theft, damage, or administrative error	SUM(shrinkage_loss)	Monthly from Gold_Fact_Inventory	Operational risk metric; high values may indicate warehouse or store-level issues
Total Units	Total quantity of units sold across all transactions	SUM(sold_units)	Monthly from Gold_Fact_Inventory	Base volume metric; used as denominator in sell-through and conversion rate calculations

Total Units Received	Total quantity of inventory received into stock from suppliers or transfers	SUM(received_stock)	Monthly from Gold_Fact_Inventory	Supply chain metric; paired with sold_units to derive closing stock reconciliation

5.5. Customer Metrics

Metric	Business Definition	SQL Logic	Grain	Notes
Active Customers	Customers with at least one completed purchase	COUNT(DISTINCT customer_id) WHERE transaction_status = 'Completed' and customer_id IS NOT NULL	Monthly from Gold_Fact_Transaction	Core base metric; denominator for all customer-level ratios
Repeat Customer Rate	Percentage of active customers who have made more than one purchase	COUNT(DISTINCT customer_id HAVING COUNT(*) > 1) / COUNT(DISTINCT customer_id)	Monthly from Gold_Fact_Transaction	Loyalty indicator; low values suggest weak retention
Purchase Frequency	Average number of completed orders per	COUNT(DISTINCT transaction_id) / COUNT(DISTINCT customer_id)	Monthly from Gold_Fact_Transaction	Engagement metric; rising trend signals stronger

	active customer			customer habits
Average Revenue per Customer	Average total revenue generated per active customer	SUM(transaction_total WHERE transaction_status = 'Completed') / COUNT(DISTINCT customer_id WHERE customer_id IS NOT NULL AND transaction_status = 'Completed')	Monthly from Gold_Fact_Transaction	Monetisation metric; used alongside Purchase Frequency to distinguish high-value vs. high-frequency segments

5.6. Marketing & Funnel Metrics

Metric	Business Definition	SQL Logic	Grain	Notes
Total Sessions	Total number of website visits recorded	COUNT(distinct session_id)	Monthly from gold_fact_clickstream	Top-of-funnel metric; baseline denominator for all session-level rates
Conversion Rate %	Percentage of sessions that result in a completed purchase	COUNT(DISTINCT session_id WHERE purchased_flag = TRUE) / COUNT(DISTINCT session_id)	Monthly from gold_fact_clickstream	Core performance KPI; primary measure of funnel

				effectivenes s
Add-to-Cart Rate %	Percentage of product page sessions where the customer added an item to cart	COUNT(DISTINCT session_id WHERE added_to_cart_flag = TRUE) / COUNT(DISTINCT session_id WHERE product_page_visited_flag = TRUE)	Monthly from gold_fact_clickstrea m	Measures purchase intent; declining trend may indicate pricing or UX friction
Bounce Rate %	Percentage of sessions where fewer than 2 pages were viewed	COUNT(DISTINCT session_id WHERE number_of_pages_viewed < 2) / COUNT(DISTINCT session_id)	Monthly from gold_fact_clickstrea m	Engagemen t quality indicator; high values suggest poor landing page relevance or slow load times
Cart Abandonmen t %	Percentage of sessions with cart activity that did not result in a purchase	(COUNT(DISTINCT session_id WHERE added_to_cart_flag = TRUE) - COUNT(DISTINCT session_id WHERE purchased_flag = TRUE)) / COUNT(DISTINCT session_id WHERE added_to_cart_flag = TRUE)	Monthly from gold_fact_clickstrea m	Friction indicator; high values suggest checkout barriers such as shipping cost or payment issues

Campaign Revenue	Total revenue from completed transactions attributed to a marketing campaign	SUM(transaction_total WHERE transaction_status = 'Completed' AND campaign_id IS NOT NULL)	Monthly from gold_fact_transaction	Attribution metric; excludes organic and unattributed transactions
Total Promotions Applied	Total number of completed transactions where a promotional code was applied	COUNT(DISTINCT transaction_id WHERE promo_id IS NOT NULL)	Monthly from gold_fact_transaction	Promotion uptake metric; use alongside Avg Revenue per Promo Transaction to assess discount effectiveness
Revenue Attributed to Campaigns %	Percentage of total completed revenue that is attributable to a marketing campaign	SUM(transaction_total WHERE transaction_status = 'Completed' AND campaign_id IS NOT NULL) / SUM(transaction_total WHERE transaction_status = 'Completed')	Monthly from gold_fact_transaction	Marketing efficiency indicator; rising share suggests campaigns are driving incremental revenue

Total Campaigns Run	Total number of distinct campaigns active within the reporting period	COUNT(DISTINCT campaign_id WHERE campaign_id IS NOT NULL)	Monthly from gold_fact_transaction	Campaign volume metric; context metric for normalising campaign revenue and attribution rates
Avg Revenue per Promo Transaction	Average revenue generated per completed transaction where a promotion was applied	SUM(line_total WHERE promo_id IS NOT NULL AND transaction_status = 'Completed') / COUNT(DISTINCT transaction_id WHERE promo_id IS NOT NULL and transaction_status = ‘Completed’)	Monthly from gold_fact_transaction	Discount effectiveness metric; compare against non-promo avg revenue to assess margin impact